```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv('/content/Zomato-data-.csv')
```

```
df.head()
```

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_in(ty
0	Jalsa	Yes	Yes	4.1/5	775	800	Bu
1	Spice Elephant	Yes	No	4.1/5	787	800	Bu
2	San Churro Cafe	Yes	No	3.8/5	918	800	Bu
3	Addhuri Udupi Bhojana	No	No	3.7/5	88	300	Bu
4	Grand Village	No	No	3.8/5	166	600	Bu

Next steps: Generate code with df | New interactive sheet

```
df.tail()
```

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed
143	Melting Melodies	No	No	3.3/5	0	100	
144	New Indraprasta	No	No	3.3/5	0	150	
145	Anna Kuteera	Yes	No	4.0/5	771	450	
146	Darbar	No	No	3.0/5	98	800	
147	Vijayalakshmi	Yes	No	3.9/5	47	200	

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 148 entries, 0 to 147
Data columns (total 7 columns):
 #   Column                       Non-Null Count  Dtype 
---  ------                       --------------  ----- 
 0   name                         148 non-null    object
 1   online_order                 148 non-null    object
 2   book_table                   148 non-null    object
 3   rate                         148 non-null    object
 4   votes                        148 non-null    int64 
 5   approx_cost(for two people)  148 non-null    int64 
 6   listed_in(type)              148 non-null    object
dtypes: int64(2), object(5)
memory usage: 8.2+ KB
```

```
df.describe()
```

	votes	approx_cost(for two people)
count	148.000000	148.000000
mean	264.810811	418.243243
std	653.676951	223.085098
min	0.000000	100.000000
25%	6.750000	200.000000
50%	43.500000	400.000000
75%	221.750000	600.000000
max	4884.000000	950.000000

```
df.shape
```

```
(148, 7)
```

```
df.columns
```

```
Index(['name', 'online_order', 'book_table', 'rate', 'votes',
       'approx_cost(for two people)', 'listed_in(type)'],
      dtype='object')
```

```
df.dtypes
```

	0
name	object
online_order	object
book_table	object
rate	object
votes	int64
approx_cost(for two people)	int64
listed_in(type)	object

dtype: object

```
df.isnull()
```

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_in(ty
0	False	False	False	False	False	False	Fa
1	False	False	False	False	False	False	Fa
2	False	False	False	False	False	False	Fa
3	False	False	False	False	False	False	Fa
4	False	False	False	False	False	False	Fa
...	...	...	...	...	...	...	
143	False	False	False	False	False	False	Fa
144	False	False	False	False	False	False	Fa
145	False	False	False	False	False	False	Fa
146	False	False	False	False	False	False	Fa
147	False	False	False	False	False	False	Fa

148 rows × 7 columns

```
df.isnull().sum
```

```
pandas.core.frame.DataFrame.sum
def sum(axis: Axis | None=0, skipna: bool=True, numeric_only: bool=False,
min_count: int=0, **kwargs)
```

/usr/local/lib/python3.12/dist-packages/pandas/core/frame.py

```
Return the sum of the values over the requested axis.

This is equivalent to the method ``numpy.sum``.

Parameters
```

```
df.fillna(0)
```

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed
0	Jalsa	Yes	Yes	4.1/5	775	800	
1	Spice Elephant	Yes	No	4.1/5	787	800	
2	San Churro Cafe	Yes	No	3.8/5	918	800	
3	Addhuri Udupi Bhojana	No	No	3.7/5	88	300	
4	Grand Village	No	No	3.8/5	166	600	
...	...	...	...	...	...	...	
143	Melting Melodies	No	No	3.3/5	0	100	
144	New Indraprasta	No	No	3.3/5	0	150	
145	Anna Kuteera	Yes	No	4.0/5	771	450	
146	Darbar	No	No	3.0/5	98	800	
147	Vijayalakshmi	Yes	No	3.9/5	47	200	

148 rows × 7 columns

```
df.dropna()
```

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed
0	Jalsa	Yes	Yes	4.1/5	775	800	
1	Spice Elephant	Yes	No	4.1/5	787	800	
2	San Churro Cafe	Yes	No	3.8/5	918	800	
3	Addhuri Udupi Bhojana	No	No	3.7/5	88	300	
4	Grand Village	No	No	3.8/5	166	600	
...	...	...	...	...	...	...	
143	Melting Melodies	No	No	3.3/5	0	100	
144	New Indraprasta	No	No	3.3/5	0	150	
145	Anna Kuteera	Yes	No	4.0/5	771	450	
146	Darbar	No	No	3.0/5	98	800	
147	Vijayalakshmi	Yes	No	3.9/5	47	200	

148 rows × 7 columns

```
df.duplicated()
```

	0
0	False
1	False
2	False
3	False
4	False
...	...
143	False
144	False
145	False
146	False
147	False

148 rows × 1 columns

dtype: bool

```
df.drop_duplicates()
```

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed
0	Jalsa	Yes	Yes	4.1/5	775	800	
1	Spice Elephant	Yes	No	4.1/5	787	800	
2	San Churro Cafe	Yes	No	3.8/5	918	800	
3	Addhuri Udupi Bhojana	No	No	3.7/5	88	300	
4	Grand Village	No	No	3.8/5	166	600	
...	...	...	...	...	...	...	
143	Melting Melodies	No	No	3.3/5	0	100	
144	New Indraprasta	No	No	3.3/5	0	150	
145	Anna Kuteera	Yes	No	4.0/5	771	450	
146	Darbar	No	No	3.0/5	98	800	
147	Vijayalakshmi	Yes	No	3.9/5	47	200	

148 rows × 7 columns

```
df.ndim
print("n dimensional:", df.ndim)
```

```
n dimensional: 2
```

```
df.shape
print("shape of the array:", df.shape)
```

```
shape of the array: (148, 7)
```

```
df.size
print("size of array:",df.size)
```

```
size of array: 1036
```

```
np.array([1,2,3])
```

```
array([1, 2, 3])
```

```
np.zeros((2,3))
```

```
array([[0., 0., 0.],
       [0., 0., 0.]])
```

```
np.ones((2,3))
```

```
array([[1., 1., 1.],
       [1., 1., 1.]])
```

```
np.arange(0,10,2)
```

```
array([0, 2, 4, 6, 8])
```

```
np.linspace(0,1,5)
```

```
array([0.  , 0.25, 0.5 , 0.75, 1.  ])
```

```
df.mean(numeric_only=True)
```

	0
votes	264.810811
approx_cost(for two people)	418.243243

dtype: float64

```
print("Median of array numeric columns only:", df.median(numeric_only=True))
```

```
Median of array numeric columns only: votes                          43.5
approx_cost(for two people)    400.0
dtype: float64
```

```
print("standard deviation value:",df.std(numeric_only=True))
```

```
standard deviation value: votes                          653.676951
approx_cost(for two people)    223.085098
dtype: float64
```

```
print("Maximum value of array:", df.max(numeric_only=True))
```

```
Maximum value of array: votes                          4884
approx_cost(for two people)     950
dtype: int64
```

```
print("minimum value of array:", df.min(numeric_only=True))
```

```
minimum value of array: votes                          0
approx_cost(for two people)    100
dtype: int64
```

```
print("sum of array:", df.sum(numeric_only=True))
```

```
sum of array: votes                          39192
approx_cost(for two people)    61900
dtype: int64
```

```
np.array([1,2,3])
```

```
array([1, 2, 3])
```

```
np.zeros((4,3))
```

```
array([[0., 0., 0.],
       [0., 0., 0.],
       [0., 0., 0.],
       [0., 0., 0.]])
```

```
np.ones((5,6))
```

```
array([[1., 1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1., 1.]])
```

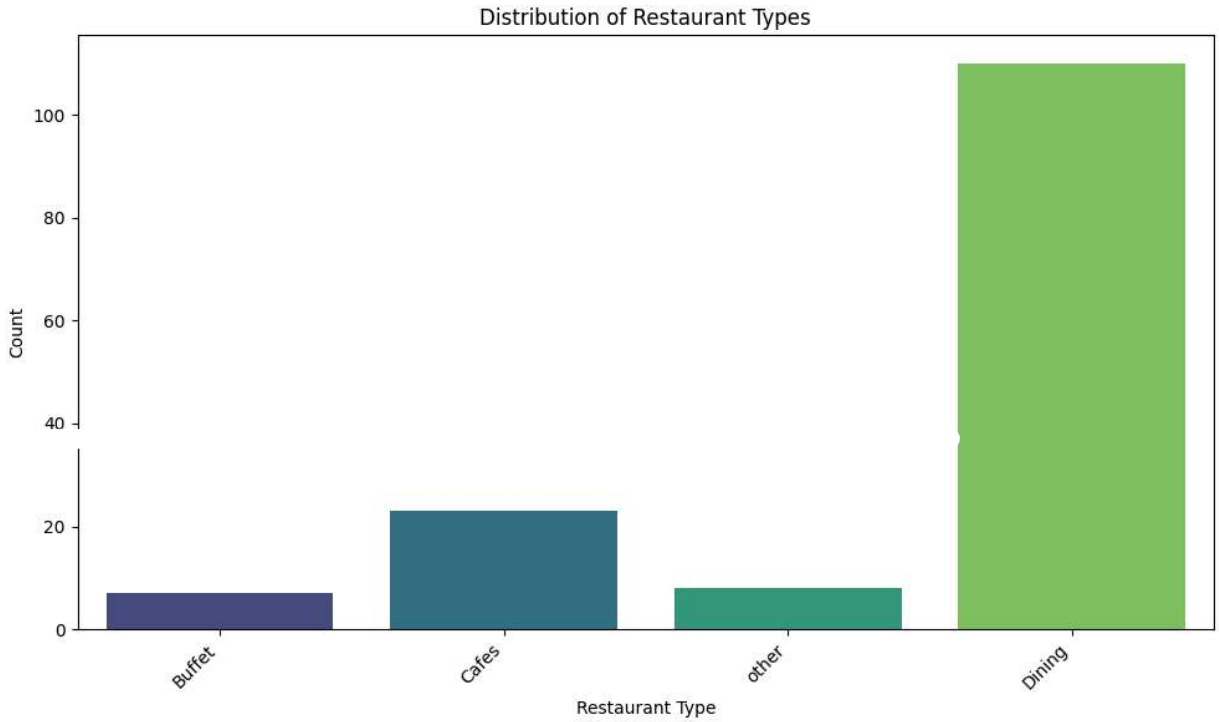
```
np.arange(0,10,2)
```

```
array([0, 2, 4, 6, 8])
```

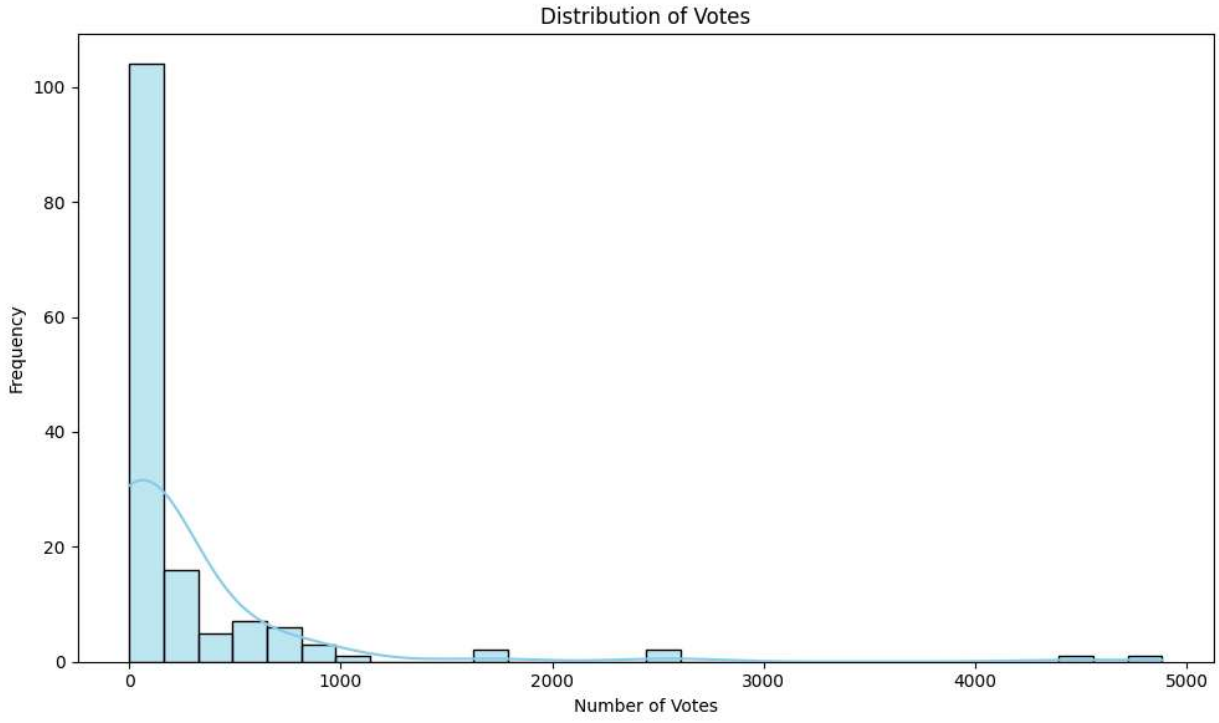
```
np.linspace(0,1,5)
```

```
array([0.  , 0.25, 0.5 , 0.75, 1.  ])
```

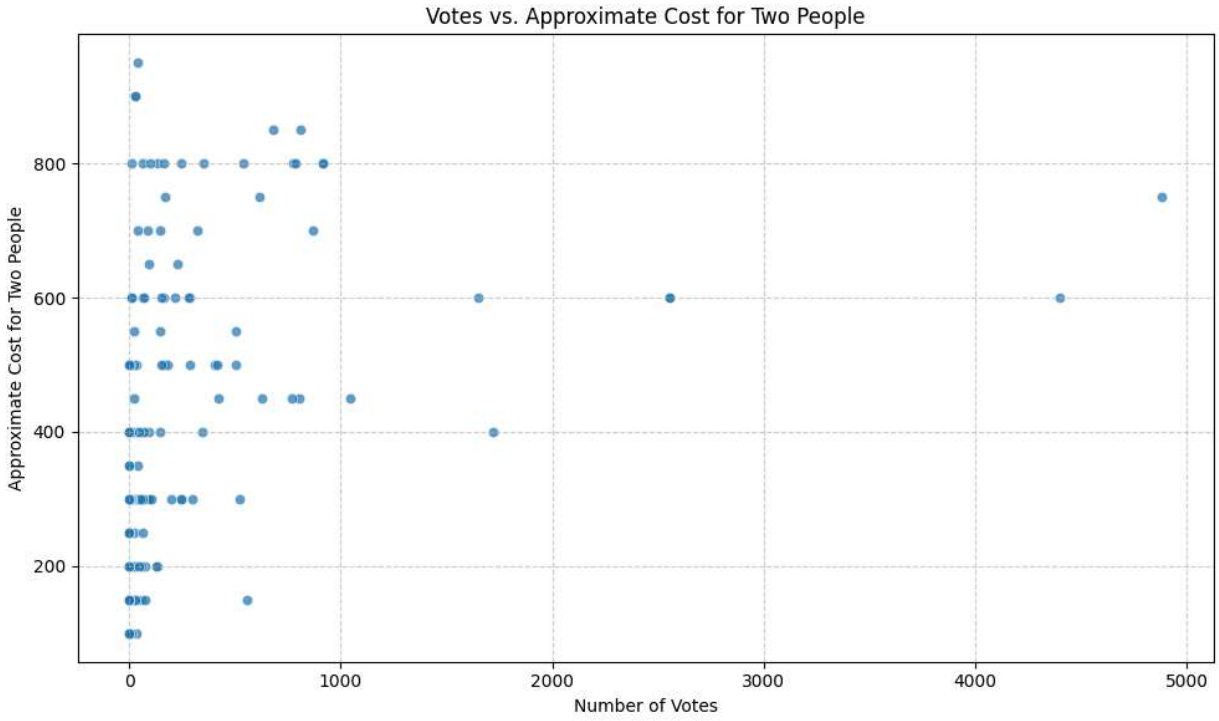
```
# Example 1: Bar plot for 'listed_in(type)' (Categorical Data)
plt.figure(figsize=(10, 6))
sns.countplot(data=df, x='listed_in(type)', hue='listed_in(type)', palette='vir
plt.title('Distribution of Restaurant Types')
plt.xlabel('Restaurant Type')
plt.ylabel('Count')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```




```
# Example 2: Histogram for 'votes' (Numerical Data)
plt.figure(figsize=(10, 6))
sns.histplot(df['votes'], bins=30, kde=True, color='skyblue')
plt.title('Distribution of Votes')
plt.xlabel('Number of Votes')
plt.ylabel('Frequency')
plt.tight_layout()
plt.show()
```



```
# Example 1: Scatter plot between 'votes' and 'approx_cost(for two people)'
plt.figure(figsize=(10, 6))
sns.scatterplot(data=df, x='votes', y='approx_cost(for two people)', alpha=0.7)
plt.title('Votes vs. Approximate Cost for Two People')
plt.xlabel('Number of Votes')
plt.ylabel('Approximate Cost for Two People')
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```




```
# Example 2: Bar plot for 'online_order' and 'approx_cost(for two people)'
plt.figure(figsize=(8, 5))
sns.barplot(data=df, x='online_order', y='approx_cost(for two people)', hue='or
plt.title('Average Approximate Cost by Online Order Availability')
plt.xlabel('Online Order Available')
plt.ylabel('Average Approximate Cost for Two People')
plt.tight_layout()
plt.show()
```




```
# Multivariate Graph 1: Scatter plot of votes vs. approximate cost, colored by
plt.figure(figsize=(12, 7))
sns.scatterplot(data=df, x='votes', y='approx_cost(for two people)', hue='onlir
plt.title('Votes vs. Approximate Cost (Colored by Online Order Availability)')
plt.xlabel('Number of Votes')
plt.ylabel('Approximate Cost for Two People')
plt.grid(True, linestyle='--', alpha=0.6)
plt.legend(title='Online Order')
plt.tight_layout()
plt.show()
```




```
# Multivariate Graph 2: Grouped bar plot of average approximate cost by restaur
plt.figure(figsize=(14, 7))
sns.barplot(data=df, x='listed_in(type)', y='approx_cost(for two people)', hue=
plt.title('Average Approximate Cost by Restaurant Type and Table Booking Availa
plt.xlabel('Restaurant Type')
plt.ylabel('Average Approximate Cost for Two People')
plt.xticks(rotation=45, ha='right')
plt.legend(title='Book Table')
plt.tight_layout()
plt.show()
```

